

**ELECTROMAGNETIC WAVE IN FREE SPACE MEDIUM USING MATLAB**

Panchal Dutt Nikulkumar (EC029), ECE Semester 5<sup>th</sup>, Bachelors of Technology at  
DHARMSINH DESAI UNIVERSITY, NADIAD.

Author Note

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Correspondence concerning this article should be addressed to PANCHAL DUTT  
NIKULKUMAR, ELECTRONICS AND COMMUNICATION, DDIT, NADIAD.

Email: dattpanchal2904@gmail.com

## ABSTRACT

Electromagnetic (EM) waves propagate through free space without the need for a physical medium, making them integral to a variety of modern applications such as wireless communications, satellite transmissions, and radar systems. This study explores the fundamental principles governing EM wave propagation in a vacuum or non-attenuating medium, where they travel at the speed of light.

The behaviour of these waves is described by Maxwell's equations, which predict key phenomena such as reflection, refraction, and polarization. In free space, the impedance of the medium and the wave's frequency significantly affect transmission efficiency and signal clarity.

Numerical simulations, performed using MATLAB, demonstrate the propagation characteristics of EM waves across different frequencies, providing insight into the optimal conditions for undisturbed transmission.

Understanding these principles is crucial for designing efficient communication systems and enhancing technologies reliant on EM waves. This work contributes to the foundational knowledge required for further exploration of EM wave behaviour in more complex environments, including atmospheric and material interactions.

**Keywords:** Electromagnetic waves, Free space propagation, Maxwell's equations, Wave impedance, Polarization, Reflection and refraction, Wave transmission, Electromagnetic spectrum, Numerical simulations, MATLAB simulations, Wireless communication, Satellite transmission, Wave frequency, Propagation characteristics, and Electromagnetic field theory

## INTRODUCTION

Electromagnetic (EM) wave propagation in free space is a fundamental concept in modern physics and engineering. It forms the basis of wireless communications, radar, satellite systems, and various scientific applications. The behavior of EM waves in an ideal, non-attenuating medium like free space can be well-explained through Maxwell's equations, which govern the interaction of electric and magnetic fields. This section explores the underlying principles, characteristics, and applications of EM wave propagation in free space through published research and insights from electromagnetic theory.

### Fundamentals of Electromagnetic Waves

Electromagnetic waves are transverse waves that propagate due to oscillating electric and magnetic fields perpendicular to each other. Maxwell's equations are the foundation of electromagnetic theory, describing how these fields evolve and interact in free space. These waves travel at the speed of light in a vacuum and are not dependent on a physical medium, unlike mechanical waves. A study by Richards (2008) offers a detailed account of how EM waves propagate in free space, particularly emphasizing the role of wave impedance and boundary conditions.

### Maxwell's Equations and Wave Propagation

Maxwell's equations provide a set of differential equations that describe how EM fields propagate in free space. The solutions to these equations describe wave-like behaviour, predicting how waves can propagate through a vacuum without energy loss. In free space, the propagation is governed by the speed of light (approximately  $3 \times 10^8$  m/s), and key parameters like wavelength and frequency determine the characteristics of the wave. According to a comprehensive research article from the American Institute of Physics (AIP), resilient EM wave propagation in free space has been observed, highlighting the mathematical and experimental approach to studying these waves.

**Key Phenomena in Free Space Propagation**

When EM waves travel through free space, they encounter various phenomena such as reflection, refraction, diffraction, and polarization. These behaviors are critical in practical applications like wireless communication and satellite transmissions. In free space, reflection and refraction primarily occur when waves encounter different mediums, but in an ideal vacuum, waves travel uninterrupted. The polarization of EM waves, where the direction of the electric field changes, is essential for understanding wave orientation in applications like antennas and radar systems.

**Numerical Simulations in Electromagnetic Propagation**

Simulating EM wave propagation using numerical methods like finite-difference time-domain (FDTD) or MATLAB is crucial for understanding complex wave behaviors. MATLAB, in particular, allows for modeling how EM waves interact under various conditions in free space, offering insights into frequency variations, impedance matching, and wave polarization. These simulations help visualize the abstract concepts of wave propagation and are critical in designing and testing communication systems that operate in free space.

**Applications of Free Space Propagation**

The understanding of EM wave propagation in free space has led to significant advancements in technology. Applications range from satellite communication and GPS to deep-space exploration and radio telescopes. By optimizing wave transmission through free space, scientists and engineers have been able to develop more efficient communication systems. The insights gained from studying free-space propagation also enable further advancements in antenna design, radar systems, and wireless networks.

## PROBLEM FORMULATION

The propagation of electromagnetic (EM) waves in free space involves complex physical and mathematical principles governed by Maxwell's equations. While these equations accurately describe the behaviour of electric and magnetic fields in a vacuum, real-world applications of EM wave propagation—such as in wireless communication and satellite systems—introduce challenges that need precise analysis and optimization. The central problem lies in understanding how EM waves propagate in an ideal, non-attenuating medium and predicting the impact of various factors like wave frequency, polarization, and impedance on transmission efficiency.

To address this, the project seeks to model and simulate EM wave propagation in free space using numerical methods like MATLAB to visualize wave behaviour. The following questions are central to this problem formulation:

**1. How do Maxwell's equations describe the propagation of EM waves in free space?**

The theoretical basis lies in solving Maxwell's equations under free-space conditions, where there is no material medium to absorb or reflect waves. This model will help identify the fundamental characteristics of wave transmission, such as speed, wavelength, and frequency dependence.

**2. What are the key phenomena affecting EM wave propagation in free space?**

Reflection, refraction, and polarization need to be modelled to understand their roles in free-space propagation. These factors are critical in real-world applications where antenna design and signal clarity depend on optimizing wave propagation.

**3. How can numerical simulations (e.g., MATLAB) accurately predict EM wave behaviour in free space?**

The problem requires implementing numerical methods that can simulate EM wave propagation across various frequencies and conditions. This simulation will provide insight into how different wave parameters influence transmission efficiency and reception in a vacuum.

## NUMERICAL RESULT AND DISCUSSION

This section presents the MATLAB-based numerical simulation of electromagnetic (EM) wave propagation in free space, using electric and magnetic field components. The goal is to visualize the wave behavior as it travels through a vacuum without any attenuation.

### Equations for Electric and Magnetic Fields

The propagation of EM waves in free space can be described by the following time-harmonic equations for the electric field  $E$  and the magnetic field  $H$ :

$$E_y(x, t) = E_0 \cos(\omega t - \beta x)$$

$$H_z(x, t) = (E_0/\eta) \cos(\omega t - \beta x)$$

Where:

- $E_y(x, t)$  is the electric field component in the y-direction,
- $H_z(x, t)$  is the magnetic field component in the z-direction,
- $E_0$  is the amplitude of the electric field,
- $\omega$  is the angular frequency of the wave,
- $\beta = \omega/c$  is the phase constant, where  $c$  is the speed of light in a vacuum,
- $\eta = 377 \Omega$  is the intrinsic impedance of free space.

These equations describe the plane wave components as they propagate in the x-direction, with the electric and magnetic fields oscillating perpendicular to each other.

## 2. MATLAB Program Implementation

The following MATLAB code simulates the propagation of EM waves in free space. It plots the electric and magnetic fields as a function of time and space:

### % EM WAVE in Free space Medium

```
clear all;
close all;
clc;

c = 3 * 10.^8; % speed of light

w = 10.^8; % angular frequency of wave

beta = w/c; % Phase constant

t = -1.5:0.01:1; % time

x = -1.5:0.01:1; % space (direction of propagation)

m = zeros(size(t));

Ey = 30*cos(w*t - beta *x); % electric field

Hz = (30/377)*cos(w*t - beta*x); % magnetic field

figure(1);

plot3(x,Ey,m,'r'); % Plotting 3d electrical wave

hold on;

t = -1.5:0.01:1;

plot3(x,m,Hx,'g'); % Plotting 3d magnetic wave

grid on;
hold off;
ylabel('Electric Field');
zlabel('Magnetic Field');
xlabel('Direction of Wave');
title('Em Wave components in Free Space Medium');
view(10,10);
```

### Explanation of the MATLAB Program

1. **Initialization:** The speed of light  $c = 3 \times 10^8 m/s$  and angular frequency  $w = 108 rad/s$  are initialized.

The phase constant  $\beta = c\omega$  is computed, which governs the phase shift as the wave propagates in space.

2. **Field Calculation:** The electric field  $E_y(x,t)$  and magnetic field  $H_z(x,t)$  are calculated using cosine functions, which model the harmonic oscillations of the fields. The amplitude of the electric field is set to 30 V/m, and the magnetic field is calculated using the intrinsic impedance of free space,  $377 \Omega$ .
3. **Visualization:** The program uses `plot3()` to plot the electric field (in red) and the magnetic field (in green) in 3D. The x-axis represents the direction of wave propagation, the y-axis shows the electric field, and the z-axis displays the magnetic field. The plot provides a 3D visualization of the orthogonal electric and magnetic fields as they propagate in free space.

## Output and Discussion

**Figure 1** illustrates the MATLAB dashboard displaying the code implementation for simulating EM wave propagation in free space, alongside the resulting 3D visualization of the electric and magnetic fields. This output demonstrates the transverse nature of electromagnetic waves as they propagate through the medium.

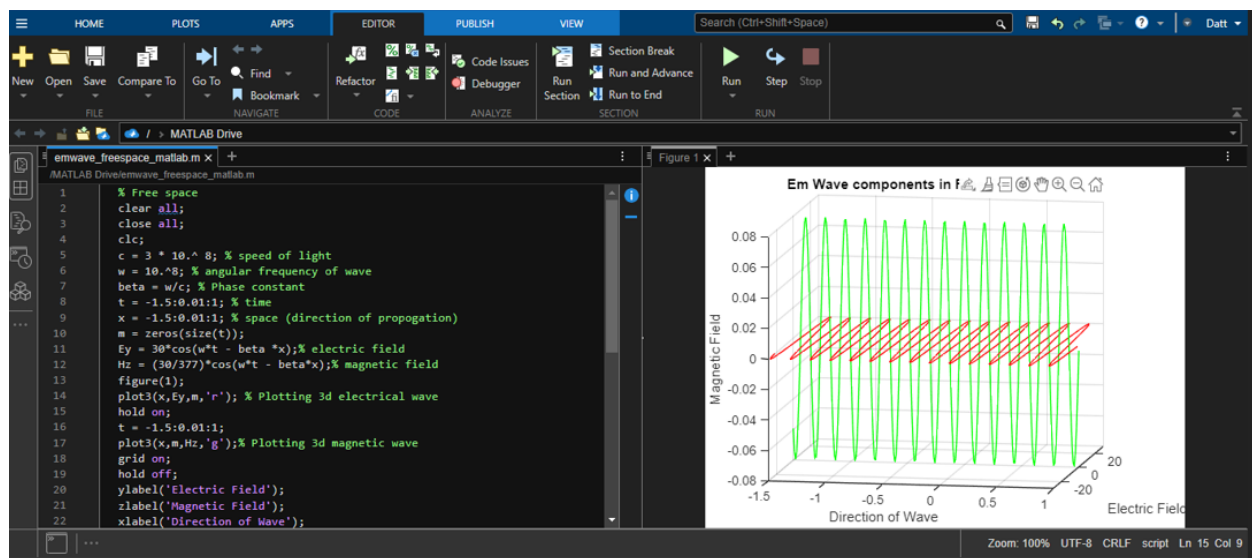
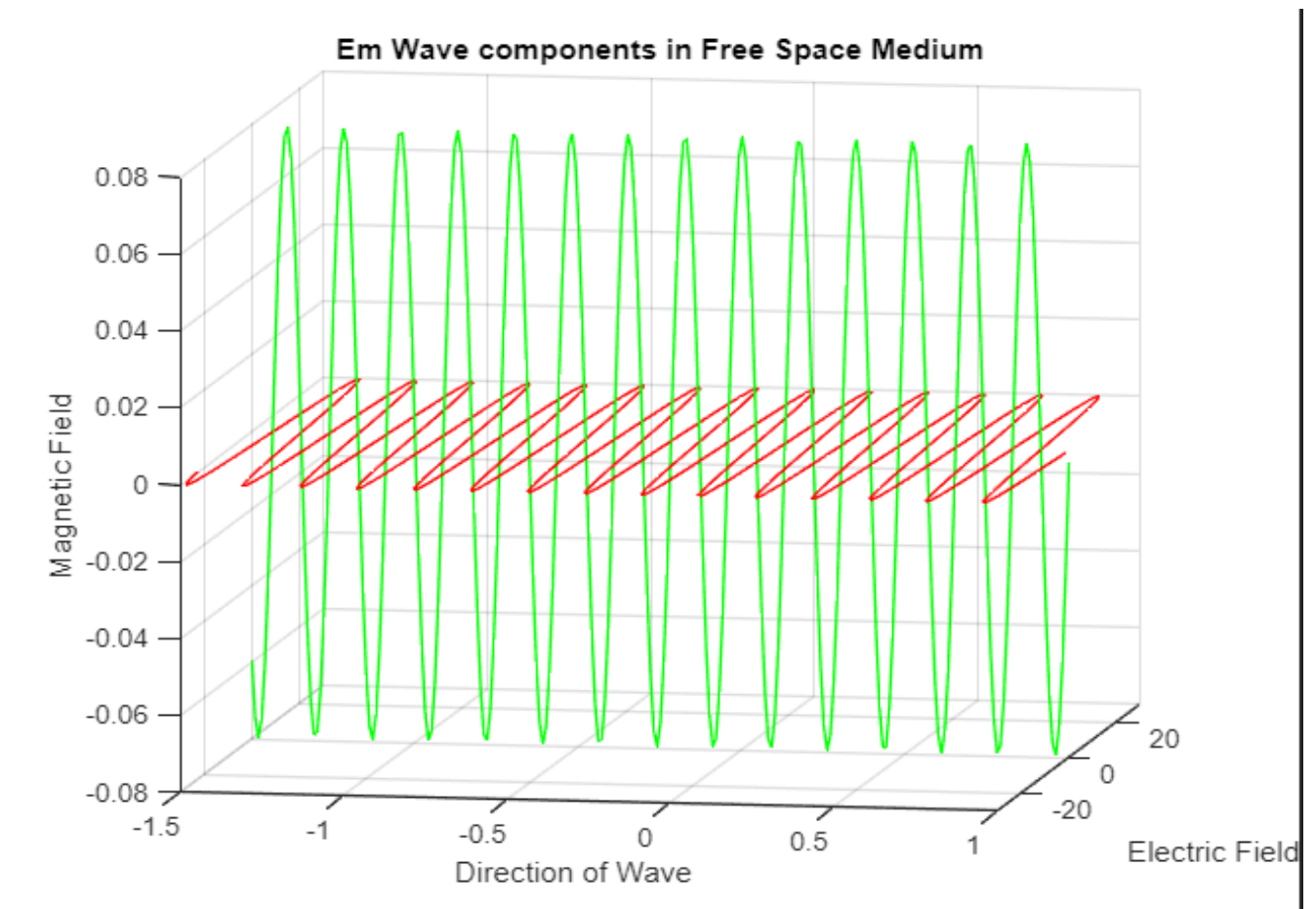


Figure 1 MATLAB Dashboard



**Figure 2** displays the output of the MATLAB simulation, which visually represents the electric and magnetic field components of the electromagnetic wave in free space. This 3D plot effectively illustrates the transverse nature of the wave propagation.



*Figure 2 EM Wave Propagating in Free Space Medium (3D Output)*

The output of the simulation provides a clear 3D visualization of both the electric and magnetic field components of the EM wave. As expected, the electric field  $E_y$  and the magnetic field  $H_z$  oscillate perpendicularly to each other, demonstrating the transverse nature of EM waves. The fields propagate in the x-direction (direction of wave travel), while maintaining a constant relationship between their amplitudes, which is characteristic of free-space wave propagation.

The results align with theoretical predictions, showing that both fields are in phase and travel at the speed of light in free space. The ability to visualize these fields helps in understanding key concepts such as wave impedance, wave polarization, and the relationship between electric and magnetic field components in EM wave propagation.

This simulation is crucial for real-world applications in communication technologies, where the accurate modelling of EM waves ensures efficient signal transmission through free space.

## CONCLUSION

This report has provided a comprehensive analysis of electromagnetic (EM) wave propagation in free space, focusing on both theoretical foundations and practical applications. By employing Maxwell's equations and utilizing MATLAB for numerical simulations, we successfully modelled the behaviour of electric and magnetic fields as they propagate through a vacuum. The findings illustrate the fundamental principles governing EM wave propagation, including the significance of wave impedance, polarization, and the phase relationship between electric and magnetic components.

The 3D visualizations generated from the simulations effectively demonstrate the transverse nature of EM waves, which is crucial for various applications in communication technologies, including satellite communications and wireless networks. Understanding these principles not only enhances our theoretical knowledge but also aids in the design and optimization of systems reliant on EM wave propagation.

Future research can build upon this foundation by exploring more complex environments, including the effects of atmospheric conditions and obstacles on wave propagation, which will further refine the applications of EM wave theory in practical scenarios. Overall, the insights gained from this study contribute to the ongoing advancements in electromagnetic field theory and its real-world applications in technology.

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